

Exact Stationary Solutions for Variants of the Smoluchowski Equation via Combinatorial Sequences

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Introduction and Motivation

The **Smoluchowski coagulation** equation is a fundamental model describing the time evolution of the number density $u(t, l)$ of particles of mass l at time t . The continuous coagulation process is driven by a symmetric kernel $K(i, j)$, which dictates the rate at which a particle of mass i coalesces with a particle of mass j :

$$\frac{d}{dt}u(t, l) = \sum_{i=1}^{l-1} K(i, l-i)u(t, i)u(t, l-i) - 2 \sum_{i=1}^{\infty} K(l, i)u(t, i)u(t, l)$$

To simplify the notation, let's call the right-hand side of the **Discrete Smoluchowski Equation** $\text{DSE}(K, u, l)$.

Next, we present modifications whose solutions are related to the following classical combinatorial sequence: Catalan Numbers, Schroder Polynomials and Motzkin Numbers.

I. Monomer Injection $\rightarrow$ Catalan Numbers

In this first scenario, we assume that *particles of mass 1 are continuously injected into the system at a constant rate $f(0)$* , so the equation became:

$$\frac{d}{dt}u(t, l) = \text{DSE}(K, u, l) + f(0)\delta_{l,1}$$

For any multiplicative kernel $K(i, j) = q(i)q(j)$, we can make the substitution $v_i = q(i)u_i$. Searching for the steady state normalized by the total mass scaling $w_l = v_l / (2 \sum_{i=1}^{\infty} v_i)$, we reach the following recurrence:

$$w_l = \sum_{i=1}^{l-1} w_i w_{l-i}$$

Which is the classic **Catalan numbers** sequence, which combinatorially represents random walks that do not cross below the horizontal axis:

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}$$

Using this, and considering the injection rate $f(0)$, we obtain the following general solution:

$$u(t, l) = u_l = \frac{2\sqrt{f(0)}}{q(l)} \left(\frac{1}{4}\right)^l \binom{2l-2}{l-1} \frac{1}{l}$$

II. Aggregate Injection $\rightarrow$ Schröder Numbers

Alternatively, we consider the case where monomers are injected directly into already existing aggregates of mass l at a rate $f(l)u(t, l)$, forcing a continuous mass shift up through successive layers, so the equation becomes:

$$\frac{d}{dt}u(t, l) = \text{DSE}(K, u, l) + f(l-1)u(t, l-1) - f(l)u(t, l) + f(0)\delta_{l,1}$$

where $u(t, 0) \equiv 0$.

For the specific kernel class $K(i, j) = f(i)f(j)$, which depends entirely on the injection rate, we can make the substitution $v_i = f(i)u_i$. Defining the boundary conservation factor $C = 1 + 2 \sum_{i=1}^{\infty} v_i$, we reach the following recurrence:

$$v_l = \frac{1}{C} \left(v_{l-1} + \sum_{i=1}^{l-1} v_i v_{l-i} \right)$$

Which closely resembles the classic **Schröder numbers** sequence, a natural generalization of Catalan numbers where lattice paths are allowed to move horizontally via diagonal steps that never dip below the x-axis:

$$S_n = S_{n-1} + \sum_{i=1}^{n-1} S_i S_{n-i}$$

Using this, we obtain the following general solution:

$$u(t, l) = u_l = \frac{1}{l \cdot 3^{l-1} \cdot f(l)} \sum_{i=1}^l \binom{l}{i} \binom{l+i-2}{i-1} \frac{1}{3^i}$$

III. Reversing the Perspective $\leftarrow$ Motzkin Numbers

After observing the emergence of Catalan and Schröder structures, we considered the reverse problem:

Given a combinatorial recurrence with a “convolutional term”, can one design a coagulation mechanism that produces it?

A natural candidate is the Motzkin recurrence:

$$M_n = M_{n-1} + \sum_{k=0}^{n-2} M_k M_{n-2-k}$$

To force the appearance of this structure, we proposed the following modification of problem II where *whenever two particles coagulate, the resulting particle receives an additional monomer*.

The corresponding dynamic equation becomes:

$$\frac{d}{dt}u(t, l) = \sum_{i=1}^{l-2} K(i, l-1-i)u(t, i)u(t, l-1-i) - 2 \sum_{i=1}^{\infty} K(l, i)u(t, l)u(t, i) + f(l-1)u(t, l-1) - f(l)u(t, l) + f(0)\delta_{l,1}$$

where $u(t, 0) \equiv 0$.

For the kernel $K(i, j) = f(i)f(j)$, we make the same substitution and algebraic steps from scenario II, we arrive at the following recurrence for $l > 1$:

$$v_l = \frac{1}{C} \left(v_{l-1} + \sum_{i=1}^{l-2} v_i v_{l-1-i} \right)$$

The recurrence can then be perfectly reduced to the classic Motzkin structure, yielding the exact closed-form solution:

$$u(t, l) = u_l = \frac{1}{l \cdot 3^l \cdot f(l)} \sum_{k=0}^{\lfloor (l-1)/2 \rfloor} \binom{l}{2k+1} \binom{2k+1}{k}$$

Conclusions and Future Work

- **The Emergence of Combinatorics:** The search for stationary solutions to variations of coagulation equation reveals an unexpected connection with combinatorial structures.
- **Convolutional Recurrences:** Solvable models emerge when dynamic modifications yield a convolutional structure. This opens up space to investigate other combinatorial families such as the **Narayana polynomials**.

$$\mathcal{N}_n(z) = (1+z)\mathcal{N}_{n-1}(z) + z \sum_{k=1}^{n-2} \mathcal{N}_k(z)\mathcal{N}_{n-k-1}(z)$$

- **Numerical Verification:** The code used to verify convergence in the first case is available at: <https://github.com/thicius>

References

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